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Advancements in Imaging Modalities for Post-Kidney Transplant Monitoring:

A Focus on Contrast-Enhanced Ultrasound

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Abbreviations

US: Ultrasound

CEUS: Contrast-Enhanced Ultrasound

TRAS: Transplant Renal Artery Stenosis

TRAT: Transplant Renal Artery Thrombosis

TRVT: Transplant Renal Vein Thrombosis

RI: Resistive Index

PSV: Peak Systolic Velocity

RAR: Renal-to-Iliac Artery Ratio

TTP: Time to Peak

TIC: Time-Intensity Curve

MTT: Mean Transit Time

CPI: Cortical Perfusion Index

PI: Pulsatility Index

AT: Acceleration Time

APN: Acute Pyelonephritis

AVF: Arteriovenous Fistula

PSA: Pseudoaneurysm

DGF: Delayed Graft Function

EDV: End-Diastolic Velocity

AI: Acceleration Index

AUC: Area Under the Curve

MRA: Magnetic Resonance Angiography

NCE-MRA: Non-Contrast-Enhanced Magnetic Resonance Angiography

CT: Computed Tomography

CTA: CT Angiography

MRI: Magnetic Resonance Imaging

Abstract: 200

Early and accurate detection of post-transplant complications is critical for preserving graft function and optimizing long-term outcomes. While conventional ultrasound is widely used in transplant monitoring, its limitations in assessing microvascular perfusion and early complications have prompted growing interest in Contrast-Enhanced Ultrasound (CEUS). CEUS provides real-time, non-nephrotoxic imaging of microvascular perfusion, making it well-suited for kidney transplant patients. We conducted a systematic review of 19 studies evaluating the use of CEUS in kidney transplant monitoring, focusing on its diagnostic accuracy, comparison with conventional US, and its role in detecting vascular complications, rejection, and perfusion abnormalities. CEUS demonstrated high diagnostic performance, with sensitivity up to 90% and specificity up to 95% for detecting vascular complications, such as transplant renal artery stenosis and transplant renal artery or vein thrombosis. CEUS also showed utility in the early detection of acute tubular necrosis and rejection. However, significant variability was noted in CEUS parameters across studies, largely due to differences in imaging software and methodology, highlighting the need for standardization. Despite its potential, the absence of standardized reference values for CEUS

parameters remains a barrier to widespread clinical adoption. Further research is needed to establish these benchmarks and optimize CEUS integration into routine transplant surveillance.

Keywords: Contrast-enhanced ultrasound, kidney transplantation, post-transplant complications, Vascular complications, Urological complications, Acute rejection, Graft surveillance, kidney graft monitoring, Super-resolution ultrasound.

Introduction

With over 90,000 patients on the kidney transplant waiting list, the persistent organ shortage underscores the urgent need to prolong graft survival and optimize available organs. Early detection of complications is critical, yet conventional tools often fail to identify microvascular dysfunction before irreversible injury. Advancing non-invasive, precise monitoring techniques is essential to improve outcomes. Ultrasound (US) remains the first-line imaging modality for renal transplant evaluation due to its real-time capabilities, absence of radiation, non-nephrotoxic contrast, and accessibility.¹ B-mode US assesses graft morphology, while Doppler evaluates blood flow to detect complications like transplant renal artery stenosis (TRAS) and renal vein thrombosis (RVT). However, Doppler is limited to large vessels, with reduced sensitivity for microvascular pathology.² Contrast-enhanced ultrasound (CEUS) addresses these limitations.³ Its intravascular microbubbles enable high-resolution imaging without nephrotoxicity. This review

examines CEUS's current role in kidney transplant follow-up and its value in complementing traditional US for post-transplant monitoring.

Methods

We conducted a comprehensive PubMed search to identify studies evaluating CEUS in kidney transplant monitoring. Major studies and expert consensus articles on conventional US (B-mode and Doppler) were also reviewed for comparative context. No language or date restrictions were applied. Reference lists were screened to identify additional relevant studies. Specific search terms and detailed methodology, including inclusion criteria, data extraction, and quality assessment, are described in the Supplementary Methods and summarized in Table S1 and Figure S1.

Results A total of 19 studies on the use of CEUS in kidney transplant monitoring, along with 9 studies in the state-of-the-art review of conventional US, were included. **Table 1** summarizes CEUS's key parameters definitions, measurement techniques, and clinical significance.

1. Vascular Complications

CEUS offers detailed visualization of kidney microcirculation, enabling imaging of vessels as small as 40 μm and individual microbubbles, which range between 2-10 μm , assessing vascular complications, infarction, and cortical necrosis. It allows quantification of cortical perfusion through quantitative methods like time-intensity curves (TIC) (**Figure 1**) and high-energy reperfusion techniques.⁴ Mueller-Peltzer et al. reported that CEUS achieved a sensitivity of 100% (based on 15 patients with vascular complications), specificity of 66.7%, positive

predictive value (PPV) of 71.4%, and negative predictive value (NPV) of 100% in detecting vascular complications in kidney transplants.⁵

Transplant Renal Artery Stenosis (TRAS) affects 1-23% of kidney transplant recipients.⁶ B-mode US struggles to detect vessel narrowing and intima thickening or irregularities. On Doppler US, PSV >200-300 cm/s indicates significant TRAS (>50%), along with a renal-to-iliac Artery Ratio (RAR) >2.5-3 (the threshold of RAR >3.5 is specific to native kidneys and does not apply to transplant kidneys), low RI (<0.5), turbulence, and spectral broadening beyond the stenotic segment.⁷ CEUS shows 87.5% sensitivity and 95.7% specificity, outperforming Doppler sensitivity of about 80% and specificity of 67.4%.⁷ CEUS is particularly valuable in cases with poor acoustic windows or inconclusive Doppler results, as it maximizes visualization of luminal narrowing (**Figure 2**), reduced downstream contrast enhancement, and prolonged Time-to-Peak (TTP) in both cortex and medulla, with perfusion delays seen as delayed cortical enhancement relative to unaffected areas.⁷

Transplant Renal Artery Thrombosis (TRAT) occurs in approximately 0.4% of cases, typically in the acute post operative phase. B-mode US has low sensitivity but may reveal increased echogenicity and heterogeneous graft echotexture. Doppler US can identify the absence of flow in the affected artery with monophasic or absent waveforms.^{8,9} However, overlying bandages, poor patient mobility, sonographer inexperience, and limited acoustic windows may reduce image quality leading to a false diagnosis of thrombosis when flow is present but not detected. CEUS offers high sensitivity and specificity (>92%) for confirming the diagnosis, showing a complete lack of contrast enhancement in the thrombosed artery. Segmental occlusions, very hard to diagnose by Doppler, are easily detected by CEUS. Parenchymal infarction manifests as hypo-perfused or non-perfused kidney regions (**Figure S2** and

Supplemental Cine Loop 1), with TIC showing delayed or absent enhancement in the cortex and pyramids.¹⁰

Transplant Renal Vein Thrombosis (TRVT) occurs in about 0.2% of kidney transplant patients, typically in the early postoperative phase. Conventional B mode US has limited diagnostic value. Spectral Doppler waveforms reveal absent venous flow, high resistance arterial flow characterized by increased RI, diminished or reversed arterial diastolic flow with transient, plateau, or inverted M waveforms, a spike-like systolic component, and the absence of flow in the transplant renal vein. However, many of these findings are not specific to TRVT and may also occur in severe acute rejection and acute tubular necrosis (ATN).¹¹ CEUS provides more precise diagnostic information, showing absent renal vein enhancement (“flow void sign”) and “patchy”, reduced, delayed, or absent parenchymal perfusion. CEUS is particularly useful in cases where the Doppler signal is poor due to technical factors or body habitus.⁸ Grzelak et al. reported significantly lower cortex and pyramids perfusion on CEUS (-53.8 ± 5.4 dB in the cortex and -54.8 ± 5.4 dB in the pyramids) compared to rejection or necrosis (-35.0 ± 3.5 dB and -37.0 ± 3.5 dB, respectively).¹²

Arteriovenous Fistula (AVF) is a common post-biopsy complication in kidney transplant patients, with an incidence ranging from 1.6% to 16.7%.¹³ B-mode US can detect the abnormal artery-to-vein connection. Doppler US shows high-velocity flow through the fistula and turbulent flow in nearby veins, with early venous enhancement and a characteristic “to-and-fro” pattern.^{9,14} However, these methods may be limited by poor acoustic windows or subtle findings, making detection challenging. CEUS is particularly useful in such scenarios, offering superior visualization of hyper-enhancing pseudo-nodules (fistula tract) with rapid contrast uptake and

clear delineation of the efferent vein's rapid filling (**Figure S3** and **Supplemental Cine Loop 2**), facilitating the assessment of embolization necessity.⁹

Acute Cortical Necrosis is a rare but severe condition caused by prolonged ischemia or hyperacute rejection, often leading to graft loss. B-mode US may reveal a hypoechoic or heterogeneous cortex with diminished corticomedullary differentiation, while Doppler US can detect reduced or absent cortical perfusion and elevated RI. However, both modalities have low sensitivity and specificity.^{15,16} In contrast, CEUS offers superior diagnostic accuracy, with 88–90% sensitivity and 80–87% specificity, enabling reliable detection of cortical necrosis without the need for additional imaging. CEUS highlights the Peripheral Rim Sign, characterized by enhancement of a peripheral rim of tissue without cortical enhancement (**Figure 3** and **Supplemental Cine Loop 3**), and visualizes a persistent hypovascular hypoechoic band up to 5 minutes post-contrast. Additionally, the Cortical Perfusion Index (CPI)—calculated as the ratio of Regional Blood Volume (RBV) to Mean Transit Time (MTT)—provides a quantitative measure of perfusion loss. CEUS also facilitates dynamic monitoring of disease progression and treatment response, offering a valuable tool for clinical decision-making.^{15,17,18}

Acute Tubular Necrosis (ATN) is a leading cause of delayed graft function (DGF), occurring in 13–50% of kidney transplants, primarily due to ischemia-reperfusion injury. B-Mode US findings are non-specific, typically showing mild kidney swelling with preserved corticomedullary differentiation or diffuse enlargement with loss of corticomedullary differentiation and increased parenchymal echogenicity. Doppler US may detect an elevated RI (>0.7 – 0.8), but these findings are not specific, and PSV and End-Diastolic Velocity (EDV) are often normal or slightly reduced, limiting diagnostic utility.^{19,20} These findings may also overlap with transplant renal vein thrombosis (TRVT), acute cortical necrosis, and rejection.

In contrast, CEUS is more sensitive to early perfusion abnormalities. It can detect reduced or heterogeneous cortical perfusion, delayed or prolonged enhancement, and increased TTP, often exceeding 30 seconds in affected areas compared to 10–15 seconds in unaffected tissue.²¹

Maximal enhancement intensity in the cortex (PI-c) and medulla (PI-m) is reduced, along with decreased CPI and the MTT (<5s, compared to the standard value of 5–10 s), indicating faster contrast passage in poorly perfused areas. AUC values typically drop by 20-40%, consistent with decrease in overall graft perfusion.^{22,23} Quantitative CEUS parameters provide high accuracy, with sensitivity and specificity exceeding 85% in some studies, making it a valuable tool for earlier ATN detection compared to conventional US, particularly in cases of DGF.²¹

Pseudoaneurysms are typically identified by B-mode US typically as round hypoechoic structures adjacent to the renal artery, but distinguishing them from other vascular or cystic structures can be challenging. Doppler US improves detection by revealing the characteristic "yin-yang" sign, indicative of turbulent blood flow within the pseudoaneurysm sac, along with to-and-fro flow at pseudoaneurysm neck.²⁴ However, Doppler US has limited sensitivity, particularly for small pseudoaneurysms (<0.5 cm) or low-flow pseudoaneurysms, which occur in 20-30% of cases.²⁴ In contrast, CEUS provides superior spatial and temporal resolution, enabling clear visualization of abnormal arterial wall bulging and contrast pooling within the aneurysmal sac, even in slow blood flow pseudoaneurysms (**Figure S4**). CEUS also offers a more volume assessment than Doppler US, and can guide interventions like percutaneous embolization, eliminating the need for nephrotoxic contrast agents.²⁵

2. Urological Complications

Ureteral Obstruction occurs in approximately 2-10% of kidney transplant recipients.²⁶ B-mode US effectively detects hydronephrosis caused by obstruction. Doppler US can identify secondary elevated RI due to urine outflow obstruction, with the acceleration phase either unchanged²⁷ or slightly delayed inflow and outflow patterns. Since ultrasound contrast agents are purely intravascular, they do not excrete into the urinary tract; thus, no intraluminal enhancement of the renal collecting system or ureters should occur under normal conditions. However, in patients with an indwelling nephrostomy tube, a CEUS nephrostogram can be performed. When contrast is injected via a nephrostomy tube, CEUS allows real-time, radiation-free visualization of the contrast flow from the kidney pelvis to the bladder,^{28,29} confirming obstruction or patency with diagnostic performance comparable to fluoroscopy (**Figure S5**).¹⁰

CEUS may play a role in diagnosing urinary leaks in kidney transplant recipients, especially when a nephrostomy catheter is already in place. Although CT urography remains the standard modality for identifying urinary extravasation, a CEUS nephrostogram can provide a valuable alternative by directly visualizing contrast leakage from the urinary tract. This approach involves the injection of ultrasound contrast through the nephrostomy tube, enabling real-time detection of leaks, particularly at the ureterovesical anastomosis site, which is the most common location. CEUS can be used not only for initial diagnosis but also during follow-up to confirm the resolution or persistence of the leak. Given its non-ionizing nature and the small contrast volume required, CEUS may be especially advantageous in selected post-transplant patients.³⁰

3. Immunologic Complications

Acute rejection typically occurs within the first few weeks or months post-transplant and has a cumulative incidence of 8% at 1 year post-transplant.³¹ B-Mode US findings are late and non-specific, often showing only mild graft enlargement. Doppler US frequently detects elevated RI (>0.7) and decreased diastolic flow in the interlobar and arcuate arteries, but these findings lack specificity.^{32,33} In contrast, CEUS is more sensitive for early rejection detection, revealing reduced or heterogeneous cortical perfusion with delayed or prolonged contrast enhancement. CEUS markers that are strong indicators of acute rejection include: increased TTP values (indicating delayed contrast inflow into the cortex, with normal TTP ranging from 7–14 seconds depending on the protocol), reduced wash-in and wash-out rates, and a decreased CPI, strong indicators of acute rejection.^{18,25,34}

An experimental model where Grabner et al. demonstrated that antibody-mediated CEUS targeting T lymphocytes could enable early detection of acute cellular rejection is described in Supplementary Results.

4. Infectious Complications

Acute Pyelonephritis (APN) is the most common infectious complication in the first year after kidney transplantation, occurring in approximately 10 to 20%, with abscess formation in 2 to 3% of cases.³⁵ B-mode US and Doppler US have low sensitivity ($<60\%$) for detecting APN, often missing hypoechoic or poorly perfused areas ($<60\%$).^{25,34} In contrast, CEUS accurately identifies hypo-enhancing regions in the cortex or medulla associated with infection.³⁶ CEUS identifies abscesses, which appear as well-defined non-enhancing cavities, with high sensitivity (95%) and specificity ($>98\%$).³⁷ Additionally, CEUS can monitor the response to antibiotic

therapy by evaluating changes in perfusion patterns over time, offering a dynamic and non-invasive follow-up tool.^{25,34}

5. Miscellaneous Complications

Long-term immunosuppression increases the risk of malignancy, often detected incidentally during routine follow-up. CT and MRI are the typical diagnostic tools for renal masses, but less optimal in this setting of renal dysfunction. B-mode US detects solid or cystic lesions, and Doppler US can reveal tumoral vascular patterns, but both are limited in providing a definitive diagnosis, except for simple cysts.³⁸ CEUS aids in differentiating benign from malignant lesions based on enhancement patterns. Bosniak I and II cysts, characterized by thin walls (<2 mm), sharp margins, and anechoic content (**Figure S6** and **Supplemental Cine Loop 4**), typically do not require CEUS. However, Bosniak IIF, III, and IV cysts often warrant further evaluation.³⁹ CEUS effectively identifies thickened wall or septa enhancement, subtle irregularities, and soft-tissue nodules (**Figure S7** and **Supplemental Cine Loop 5**), helping distinguish benign from suspicious features.⁴⁰ For solid masses, hypervascular patterns on CEUS may suggest malignancy (**Figure S8** and **Supplemental Cine Loop 6**), though specific CEUS vascularization patterns for solid kidney neoplasms remain under-documented in the literature.⁴¹ CEUS may assist in early detection of posttransplant lymphoproliferative disorders (PTLD), but current data remain limited to case reports. Additionally, CEUS can assess urothelial lesions, where intravascular microbubbles help differentiate soft-tissue masses from thrombi, a common differential in transplant recipients.

Postoperative fluid collections (hematomas, seromas, urinomas, lymphoceles) are common but difficult to differentiate on B-mode US, especially for iso- or hyperechoic hematomas. CEUS improves detection by 17.6%, provides precise localization and measurement, and helps assess wall thickness to determine the need for drainage.^{42,43}

The contribution of CEUS in pre- and post-biopsy evaluation is described in the Supplementary Results, such as a proposal of a standardized CEUS protocol.

Discussion

CEUS has emerged as a valuable imaging modality in kidney transplantation management. A critical advantage of CEUS is its ability to provide non-invasive, rapid bedside assessments without the risks associated with nephrotoxic contrast agents. This makes it particularly suited for kidney transplant recipients, who often have impaired renal function and require frequent imaging.³ While conventional US techniques remain integral to graft monitoring, their limitations underscore the need for advanced modalities like CEUS, which provides earlier and more accurate detection of complications, allowing for more timely and effective treatment. Routine use of CEUS has the potential to improve graft and patient outcomes. This review consolidates updated, clinically relevant insights into the utility of CEUS in kidney transplant follow-up.

Enhanced Diagnostic Precision with CEUS

CEUS excels in diagnosing vascular complications such as TRAS, TRAT, and TRVT, achieving up to 90% sensitivity and 95% specificity. Its real-time, high-resolution imaging of microvascular perfusion enables earlier detection of subtle flow abnormalities.⁹ Similarly, in

acute rejection, CEUS surpasses Doppler with a sensitivity and specificity reaching 85–90% and 95%, respectively. This facilitates earlier and appropriate treatment, such as adjustments of the immunosuppressive regimen.³³ Furthermore, CEUS aids in the detection of neoplasms, offering a safer, contrast-sparing alternative to CT or MRI, especially in patients with impaired kidney function, although additional imaging may sometimes be required for definitive diagnosis.¹⁰

Challenges and Limitations

Despite broad availability, CEUS implementation varies. In some centers, trained sonographers perform the full procedure including IV insertion and contrast administration; in others, radiologists are more involved in acquisition. Not all ultrasound machines are CEUS-capable, and some require dedicated software activation. Post-processing for quantitative analysis (e.g., mean transit time [MTT], time-to-peak [TTP]) may also require off-cart software. Adoption depends on addressing these logistical issues, such as training personnel and standardizing acquisition protocols⁴⁴.

Operator dependency remains a concern, with diagnostic quality hinging on experience and technique.^{10,25} Acoustic window limitations, due to patient habitus, bandages, or surgical clips, can reduce image quality.⁴⁵ Although the contrast agent (Lumason®/SonoVue®) is generally safe, rare allergic reactions or side effects (e.g., headache) have been reported and must be considered in transplant recipients.⁴⁶

While CEUS is highly effective, CT or MRI may be preferred in some scenarios. CT excels in confirming urine leaks or complex vascular abnormalities but poses risks due to nephrotoxicity and radiation. MRI, especially with contrast, offers detailed evaluation of masses or parenchymal changes but is limited by the risk of nephrogenic systemic fibrosis in patients with reduced renal

function. Non-contrast MRI techniques offer safer alternatives but with reduced diagnostic yield.⁴⁷

Future Perspectives

Emerging technologies such as super-resolution CEUS and quantitative perfusion analysis promise even greater diagnostic accuracy, including capillary-level resolution.^{48,49} Future research should focus on standardizing protocols and validating these innovations in clinical settings.

Conclusion

CEUS bridges a critical need in kidney transplant management, providing a non-nephrotoxic, radiation-free imaging option with real-time perfusion assessment capabilities. By enabling earlier and more precise detection of a broad spectrum of complications, CEUS has the potential to optimize patient outcomes. However, its widespread adoption is currently limited by practical challenges, including the lack of site experience, the time, effort, and expertise required to perform the procedure, and inpatient reimbursement hurdles. Future efforts should establish consistent CEUS protocols, reference norms, and diagnostic frameworks to integrate this promising technology fully into routine clinical practice.

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- David E, Del Gaudio G, Drudi FM, et al. Contrast Enhanced Ultrasound Compared with MRI and CT in the Evaluation of Post-Renal Transplant Complications. *Tomography*. 2022;8(4):1704-1715. doi:10.3390/tomography8040143
- Araújo NC, Suassuna JHR. Contrast-Enhanced Ultrasound Reveals Masked Post-Baseline Wedge Transplant Biopsy Pseudoaneurysm. *J Diagn Med Sonogr*. 2015;31(5):331-334. doi:10.1177/8756479315598523
- Chi T, Usawachintachit M, Weinstein S, et al. Contrast Enhanced Ultrasound as a Radiation-Free Alternative to Fluoroscopic Nephrostogram for Evaluating Ureteral Patency. *J Urol*. 2017;198(6):1367-1373. doi:10.1016/j.juro.2017.07.074

Conflict of Interest: None

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Figure Legends

Figure 1. Quantitative indexes of perfusion (Time-intensity curves plotting contrast enhancement intensity versus time). (A) Healthy kidney. (B) Ischemic kidney.¹⁰

Figure 2. Transplant renal artery stenosis. (A) Doppler ultrasonography demonstrating elevated peak systolic velocity (446.4cm/s) at the origin of the main renal transplant artery, associated with downstream tardus-parvus waveforms (B), suggestive of a significant stenosis. (C) CEUS image showing focal arterial narrowing just distal to the anastomosis (arrow). The minimal luminal diameter measures approximately 0.52 cm, compared to a more distal segment measuring 1.09 cm. According to NASCET criteria, this corresponds to an estimated 55% stenosis.

Figure 3. Acute cortical necrosis. (A) CEUS on day 0 post familial kidney transplantation, showing no recovery of diuresis. Doppler ultrasound revealed decreased vascularity. CEUS demonstrates well-defined non-enhancing cortical areas (arrows), with preservation of the peripheral cortical rim (cortical cortices) and the junctional cortex (suggesting a shunting pattern sparing the medulla). (B) Simultaneously acquired grayscale B-mode image for anatomical reference (with low mechanical index).

Supplemental Figures and Cine Loops

Supplemental Figures Legends

Supplemental Figure 1. Selection process pipeline.

Supplemental Figure 2. Parenchymal infarction. (A) CEUS image in a 72-year-old patient, five days post renal transplantation, with no recovery of graft function. Doppler ultrasound was non-contributory. CEUS reveals a large cortical infarction (arrows) characterized by a complete absence of enhancement. (B) Simultaneously acquired grayscale B-mode image (with low mechanical index) used for anatomical localization.

Supplemental Figure 3. Arteriovenous fistula. (A) CEUS in a 52-year-old female patient, two months after kidney transplantation. CEUS was performed as part of post-biopsy surveillance following suspicion of an arteriovenous fistula on Doppler ultrasound. Early opacification of a dilated draining vein is seen during the arterial phase (arrows), confirming the presence of an arteriovenous shunt. (B) Simultaneously acquired grayscale B-mode image for anatomical reference (using low mechanical index).

Supplemental Figure 4. Pseudoaneurysms. (A) Contrast-enhanced color Doppler image of the transplant kidney showing a small region of flow in the lower pole (arrow) in the cortical region. (B) Contrast-enhanced gray-scale image of the lower pole of the transplant kidney clearly showing the site and size of the small pseudoaneurysm (arrow). (C) Follow-up contrast-enhanced color Doppler image of the site of the previously seen pseudoaneurysm (arrow) with no evidence of flow within the sac. (D) Follow-up contrast-enhanced gray-scale image of the site of the previously seen pseudoaneurysm (arrow) with no evidence of flow within the sac, indicative of complete spontaneous thrombosis.⁵⁰

Supplemental Figure 5. CEUS nephrostograms. CEUS and fluoroscopic nephrostogram images of 60-year-old male with patent collecting system 1 day after percutaneous nephrolithotomy. (A) 1 minute after injecting ultrasound contrast agent via preplaced nephrostomy tube CEUS of bladder identified contrast agent in bladder (dashed arrow). Note

Foley balloon (solid arrow). (B) Fluoroscopic nephrostogram confirmed ureteral patency and contrast agent entering bladder (dashed arrow). **CEUS and fluoroscopic nephrostogram images of 64-year-old female with obstructed collecting system.** (C) 10 minutes after injecting ultrasound contrast agent via preplaced nephrostomy tube CEUS of bladder failed to identify any contrast agent in bladder lumen. Note Foley balloon (arrow). (D) Fluoroscopic nephrostogram confirmed persistent obstruction of left collecting system with no spontaneous flow of contrast material into ureter or bladder.²⁸

Supplemental Figure 6. Simple cyst. (A) CEUS in a 45-year-old male patient, showing a well-defined 18 mm renal cyst (arrow) incidentally detected during follow-up ultrasound. There was uncertainty on B-mode and Doppler imaging regarding the presence of a septation. The cyst shows no mural enhancement, septations, or solid components, consistent with a simple cyst, Bosniak I equivalent. (B) Simultaneously acquired grayscale B-mode image for anatomical correlation (using low mechanical index).

Supplemental Figure 7. Bosniak IV cyst. (A) Color Doppler ultrasound showing a complex renal cyst with internal echogenic components and suspected vascularization (thin arrows). (B) CEUS confirming enhancement of two solid components (thick arrows), equivalent to a Bosniak IV cyst.

Supplemental Figure 8. Tubulopapillary carcinoma. CEUS image in a 65-year-old male patient, 15 years post kidney transplantation. A mass was incidentally discovered during routine surveillance ultrasound but was not characterizable on B-mode or Doppler imaging. CEUS revealed a 25 mm hypovascular solid lesion (arrow). Percutaneous biopsy confirmed the diagnosis of a tumor, consistent with tubulopapillary carcinoma.

Supplemental Cine Loops Legends

Supplemental Cine Loop 1. Parenchymal infarction. CEUS cine loop showing a large perfusion defect in the renal graft, consistent with cortical infarction.

Supplemental Cine Loop 2. Arteriovenous fistula. CEUS cine loop showing early arterial-phase opacification of a dilated draining vein in a 52-year-old kidney transplant recipient, two months post-transplantation. The enhancement pattern confirms the presence of an arteriovenous fistula following biopsy, with rapid venous filling indicative of a high-flow arteriovenous shunt.

Supplemental Cine Loop 3. Acute cortical necrosis. CEUS cine loop acquired immediately post-transplantation (day 0). The cine demonstrates sharply demarcated cortical perfusion defects, with preserved enhancement of the peripheral cortical rim and junctional cortex, consistent with segmental cortical ischemia and shunting that spares the medulla.

Supplemental Cine Loop 4. Simple cyst. CEUS cine loop illustrating a well-defined, non-enhancing 18 mm renal cyst in a 45-year-old male patient. The cyst shows no mural enhancement, septations, or solid components, consistent with a simple cyst, Bosniak I equivalent. The grayscale B-mode image acquired simultaneously (right) confirms the anechoic content and thin walls.

Supplemental Cine Loop 5. Bosniak IV cyst. CEUS cine loop demonstrating a complex renal cyst with two enhancing intracystic solid components, equivalent to a Bosniak IV cyst. The enhancement pattern is indicative of vascularized tissue suggestive of malignancy.

Supplemental Cine Loop 6. Tubulopapillary carcinoma. CEUS cine loop in a 65-year-old kidney transplant recipient (15 years post-transplant), revealing a 25 mm hypovascular solid lesion within the graft. The lesion was not characterizable on grayscale or Doppler imaging. CEUS enhancement pattern was suggestive of a malignant tumor, later confirmed by biopsy as a tubulopapillary carcinoma.

References

1. Kasiske BL, Zeier MG, Chapman JR, et al. KDIGO clinical practice guideline for the care of kidney transplant recipients: a summary. *Kidney Int.* 2010;77(4):299-311. doi:10.1038/ki.2009.377
2. Trunz LM, Balasubramanya R. Doppler Renal Assessment, Protocols, and Interpretation. In: *StatPearls*. StatPearls Publishing; 2024. Accessed September 29, 2024. <http://www.ncbi.nlm.nih.gov/books/NBK572135/>
3. Claudon M, Cosgrove D, Albrecht T, et al. Guidelines and Good Clinical Practice Recommendations for Contrast Enhanced Ultrasound (CEUS) - Update 2008. *Ultraschall Med - Eur J Ultrasound.* 2008;29(01):28-44. doi:10.1055/s-2007-963785
4. Harvey C, Sidhu P, Bachmann Nielsen M. Contrast-Enhanced Ultrasound in Renal Transplants: Applications and Future Directions. *Ultraschall Med - Eur J Ultrasound.* 2013;34(04):319-321. doi:10.1055/s-0033-1350138
5. Mueller-Peltzer K, Rübenhaller J, Fischereder M, Habicht A, Reiser M, Clevert DA. The diagnostic value of contrast-enhanced ultrasound (CEUS) as a new technique for imaging of vascular complications in renal transplants compared to standard imaging modalities. Jünger M, Krüger-Genge A, Jung F, eds. *Clin Hemorheol Microcirc.* 2017;67(3-4):407-413. doi:10.3233/CH-179221
6. Baird DP, Williams J, Petrie MC, Smith JR. Transplant Renal Artery Stenosis. *Kidney Int Rep.* 2020;5(12):2399-2402. doi:10.1016/j.ekir.2020.09.041

7. Adani GL, Como G, Bonato F, et al. Detection of transplant renal artery stenosis with contrast-enhanced ultrasound. *Radiol Case Rep*. 2018;13(4):890-894. doi:10.1016/j.radcr.2018.06.003
8. Grzelak P, Kurnatowska I, Nowicki M, et al. Perfusion Disturbances of Kidney Graft Parenchyma Evaluated with Contrast-Enhanced Ultrasonography in the Immediate Period following Kidney Transplantation. *Nephron Clin Pract*. 2013;124(3-4):173-178. doi:10.1159/000355633
9. Granata A, Clementi S, Londrino F, et al. Renal transplant vascular complications: the role of Doppler ultrasound. *J Ultrasound*. 2015;18(2):101-107. doi:10.1007/s40477-014-0085-6
10. David E, Del Gaudio G, Drudi FM, et al. Contrast Enhanced Ultrasound Compared with MRI and CT in the Evaluation of Post-Renal Transplant Complications. *Tomography*. 2022;8(4):1704-1715. doi:10.3390/tomography8040143
11. Khaled El Zorkany. Transplant Renal Vein Thrombosis. *Exp Clin Transplant*. 15(2).
12. Grzelak P, Kurnatowska I, Sapieha M, et al. Disturbances of Kidney Graft Perfusion as Indicators of Acute Renal Vein Thrombosis in Contrast-Enhanced Ultrasonography. *Transplant Proc*. 2011;43(8):3018-3020. doi:10.1016/j.transproceed.2011.08.067
13. Schwarz A, Hiss M, Gwinner W, Becker T, Haller H, Keberle M. Course and relevance of arteriovenous fistulas after renal transplant biopsies. *Am J Transplant Off J Am Soc Transplant Am Soc Transpl Surg*. 2008;8(4):826-831. doi:10.1111/j.1600-6143.2008.02160.x
14. Chen MC, Tsai WL, Tsai IC, et al. Arteriovenous Fistula and Graft Evaluation in Hemodialysis Patients Using MDCT: A Primer. *Am J Roentgenol*. 2010;194(3):838-847. doi:10.2214/AJR.09.3357
15. Fernandez C, Ripolles T, Martinez M, Blay J, Pallardó L, Gavela E. Diagnosis of Acute Cortical Necrosis in Renal Transplantation by Contrast-Enhanced Ultrasound: a Preliminary Experience. *Ultraschall Med - Eur J Ultrasound*. 2012;34(04):340-344. doi:10.1055/s-0032-1313007
16. Spiesecke P, Münch F, Fischer T, Hamm B, Lerchbaumer MH. Multiparametric ultrasound findings in acute kidney failure due to rare renal cortical necrosis. *Sci Rep*. 2021;11(1):2060. doi:10.1038/s41598-021-81690-x
17. Helck A, Sommer WH, Wessely M, Notohamiprodjo M, Reiser M, Clevert DA. Benefit of contrast enhanced ultrasound for detection of ischaemic lesions and arterio venous fistulas in renal transplants – A feasibility study. *Clin Hemorheol Microcirc*. 2011;48(1-3):149-160. doi:10.3233/CH-2011-1398

18. Araújo NC, Suassuna JHR. Time-intensity curve analysis of contrast-enhanced ultrasound is unable to differentiate renal dysfunction in the early post-transplant period - a prospective study. *BMC Nephrol.* 2018;19(1):351. doi:10.1186/s12882-018-1158-0
19. Naesens M, Heylen L, Lerut E, et al. Intrarenal Resistive Index after Renal Transplantation. *N Engl J Med.* 2013;369(19):1797-1806. doi:10.1056/NEJMoa1301064
20. Cano H, Castañeda DA, Patiño N, et al. Resistance Index Measured by Doppler Ultrasound as a Predictor of Graft Function After Kidney Transplantation. *Transplant Proc.* 2014;46(9):2972-2974. doi:10.1016/j.transproceed.2014.07.001
21. Dong S, Zhang B, Chen Z, et al. Assessment of renal quality with quantitative contrast-enhanced ultrasound (CEUS) for differentiating kidney histopathology before procurement. *Int J Med Sci.* 2024;21(1):19-26. doi:10.7150/ijms.88147
22. Song J, Yao Y, He Y, Lin S, Pan S, Zhong M. Contrast-Enhanced Ultrasonography Value for Early Prediction of Delayed Graft Function in Renal Transplantation Patients. *J Ultrasound Med.* 2023;42(1):201-210. doi:10.1002/jum.16010
23. Bellos I, Perrea DN, Kontzoglou K. Renal resistive index as a predictive factor of delayed graft function: A meta-analysis. *Transplant Rev.* 2019;33(3):145-153. doi:10.1016/j.trre.2019.03.003
24. Ngo TC, Lee JJ, Gonzalgo ML. Renal pseudoaneurysm: an overview. *Nat Rev Urol.* 2010;7(11):619-625. doi:10.1038/nrurol.2010.163
25. Como G, Re JD, Adani GL, Zuiani C, Girometti R. Role for contrast-enhanced ultrasound in assessing complications after kidney transplant. *World J Radiol.* 2020;12(8):156-171. doi:10.4329/wjr.v12.i8.156
26. Kumar S, Ameli-Renani S, Hakim A, Jeon JH, Shrivastava S, Patel U. Ureteral obstruction following renal transplantation: causes, diagnosis and management. *Br J Radiol.* 2014;87(1044):20140169. doi:10.1259/bjr.20140169
27. Department of Radiology and Diagnostic Imaging, Norbert Barlicki Memorial Teaching Hospital No. 1 of The Medical University of Lodz, Poland, Ustyniak S, Kurnatowska I, et al. Is contrast-enhanced ultrasonography (CEUS) useful in the assessment of obstructive uropathy-induced perfusion disorders? *J Ultrason.* 2021;21(87):348-352. doi:10.15557/JoU.2021.0057
28. Chi T, Usawachintachit M, Weinstein S, et al. Contrast Enhanced Ultrasound as a Radiation-Free Alternative to Fluoroscopic Nephrostogram for Evaluating Ureteral Patency. *J Urol.* 2017;198(6):1367-1373. doi:10.1016/j.juro.2017.07.074

29. Daneshi M, Yusuf GT, Fang C, Sellars ME, Huang DY, Sidhu PS. Contrast-enhanced ultrasound (CEUS) nephrostogram: utility and accuracy as an alternative to fluoroscopic imaging of the urinary tract. *Clin Radiol*. 2019;74(2):167.e9-167.e16. doi:10.1016/j.crad.2018.10.004
30. Englesbe MJ, Dubay DA, Gillespie BW, et al. Risk factors for urinary complications after renal transplantation. *Am J Transplant Off J Am Soc Transplant Am Soc Transpl Surg*. 2007;7(6):1536-1541. doi:10.1111/j.1600-6143.2007.01790.x
31. Goutaudier V, Danger R, Catar RA, et al. Evaluation of non-invasive biomarkers of kidney allograft rejection in a prospective multicenter unselected cohort study (EU-TRAIN). *Kidney Int*. 2024;106(5):943-960. doi:10.1016/j.kint.2024.07.027
32. Hanssen O, Erpicum P, Lovinfosse P, et al. Non-invasive approaches in the diagnosis of acute rejection in kidney transplant recipients. Part I. *In vivo* imaging methods. *Clin Kidney J*. Published online July 28, 2016:sfw062. doi:10.1093/ckj/sfw062
33. Zhou Q, Yu Y, Qin W, et al. Current Status of Ultrasound in Acute Rejection After Renal Transplantation: A Review with a Focus on Contrast-Enhanced Ultrasound. *Ann Transplant*. 2021;26. doi:10.12659/AOT.929729
34. Granata A, Di Nicolò P, Scarfia VR, et al. Renal transplantation parenchymal complications: what Doppler ultrasound can and cannot do. *J Ultrasound*. 2015;18(2):109-116. doi:10.1007/s40477-014-0118-1
35. Pacaud M, Colas L, Kerleau C, et al. Impact of Late and Recurrent Acute Graft Pyelonephritis on Long-Term Kidney Graft Outcomes. *Front Immunol*. 2022;13:824425. doi:10.3389/fimmu.2022.824425
36. Rinaldo C, Grimaldi D, Di Serafino M, et al. An update on pyelonephritis: role of contrast enhancement ultrasound (CEUS). *J Ultrasound*. 2022;26(2):333-342. doi:10.1007/s40477-022-00733-9
37. Granata A, Andrulli S, Fiorini F, et al. Diagnosis of acute pyelonephritis by contrast-enhanced ultrasonography in kidney transplant patients. *Nephrol Dial Transplant*. 2011;26(2):715-720. doi:10.1093/ndt/gfq417
38. Herms E, Weirich G, Maurer T, et al. Ultrasound-based "CEUS-Bosniak" classification for cystic renal lesions: an 8-year clinical experience. *World J Urol*. 2022;41(3):679-685. doi:10.1007/s00345-022-04094-0
39. Sidhu P, Cantisani V, Dietrich C, et al. The EFSUMB Guidelines and Recommendations for the Clinical Practice of Contrast-Enhanced Ultrasound (CEUS) in Non-Hepatic Applications: Update 2017 (Short Version). *Ultraschall Med - Eur J Ultrasound*. 2018;39(02):154-180. doi:10.1055/s-0044-101254

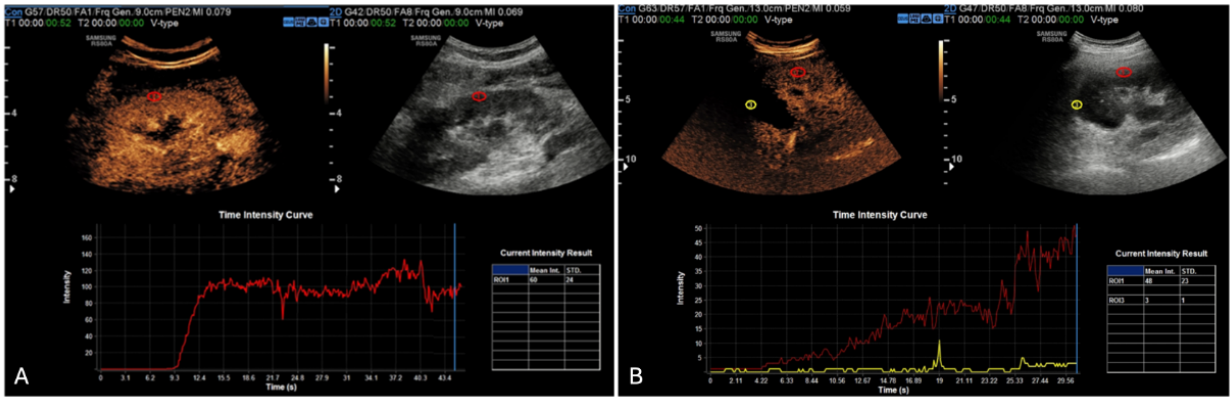
40. Cantisani V, Bertolotto M, Clevert DA, et al. EFSUMB 2020 Proposal for a Contrast-Enhanced Ultrasound-Adapted Bosniak Cyst Categorization – Position Statement. *Ultraschall Med - Eur J Ultrasound*. 2021;42(02):154-166. doi:10.1055/a-1300-1727
41. Harvey CJ, Alsafi A, Kuzmich S, et al. Role of US Contrast Agents in the Assessment of Indeterminate Solid and Cystic Lesions in Native and Transplant Kidneys. *RadioGraphics*. 2015;35(5):1419-1430. doi:10.1148/rg.2015140222
42. Fananapazir G, Rao R, Corwin MT, Naderi S, Santhanakrishnan C, Troppmann C. Sonographic Evaluation of Clinically Significant Perigraft Hematomas in Kidney Transplant Recipients. *Am J Roentgenol*. 2015;205(4):802-806. doi:10.2214/AJR.15.14426
43. Kazmierski BJ, Sharbidre KG, Robbin ML, Grant EG. CONTRAST-ENHANCED Ultrasound for the Evaluation of Renal Transplants. *J Ultrasound Med*. 2020;39(12):2457-2468. doi:10.1002/jum.15339
44. Strom JB, Appis A, Barr RG, et al. Multi-societal expert consensus statement on the safe administration of ultrasound contrast agents. *Echo Res Pract*. 2025;12(1):4. doi:10.1186/s44156-024-00068-7
45. Fetzer DT, Vijay K, Caserta MP, Patterson-Lachowicz A, Dahiya N, Rodgers SK. Artifacts and Technical Considerations at Contrast-enhanced US. *RadioGraphics*. 2023;43(2):e220093. doi:10.1148/rg.220093
46. Jung EM, Stroszczyński C, Jung F. Contrast enhanced ultrasonography (CEUS) to detect abdominal microcirculatory disorders in severe cases of COVID-19 infection: First experience. *Clin Hemorheol Microcirc*. 2020;74(4):353-361. doi:10.3233/CH-209003
47. Schutter R, Lantinga VA, Borra RJH, Moers C. MRI for diagnosis of post-renal transplant complications: current state-of-the-art and future perspectives. *Magn Reson Mater Phys Biol Med*. 2020;33(1):49-61. doi:10.1007/s10334-019-00813-8
48. Emanuel AL, Meijer RI, Van Poelgeest E, Spoor P, Serné EH, Eringa EC. Contrast-enhanced ultrasound for quantification of tissue perfusion in humans. *Microcirculation*. 2020;27(1):e12588. doi:10.1111/micc.12588
49. Bodard S, Denis L, Chabouh G, et al. Visualization of Renal Glomeruli in Human Native Kidneys With Sensing Ultrasound Localization Microscopy. *Invest Radiol*. Published online January 13, 2024. doi:10.1097/RLI.0000000000001061
50. Araújo NC, Suassuna JHR. Contrast-Enhanced Ultrasound Reveals Masked Post-Baseline Wedge Transplant Biopsy Pseudoaneurysm. *J Diagn Med Sonogr*. 2015;31(5):331-334. doi:10.1177/8756479315598523

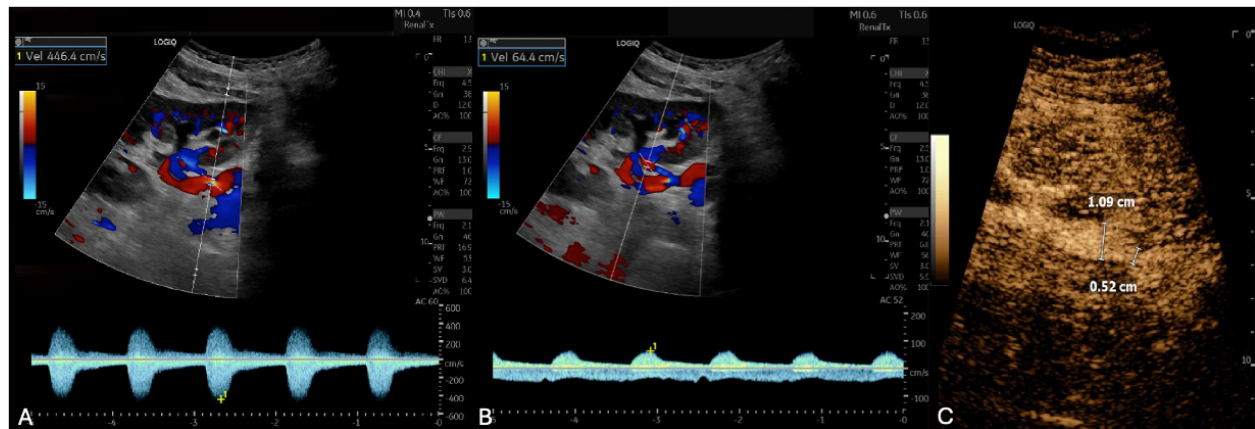
Table 1. Key Metrics for Assessing Renal Transplant Health Using CEUS

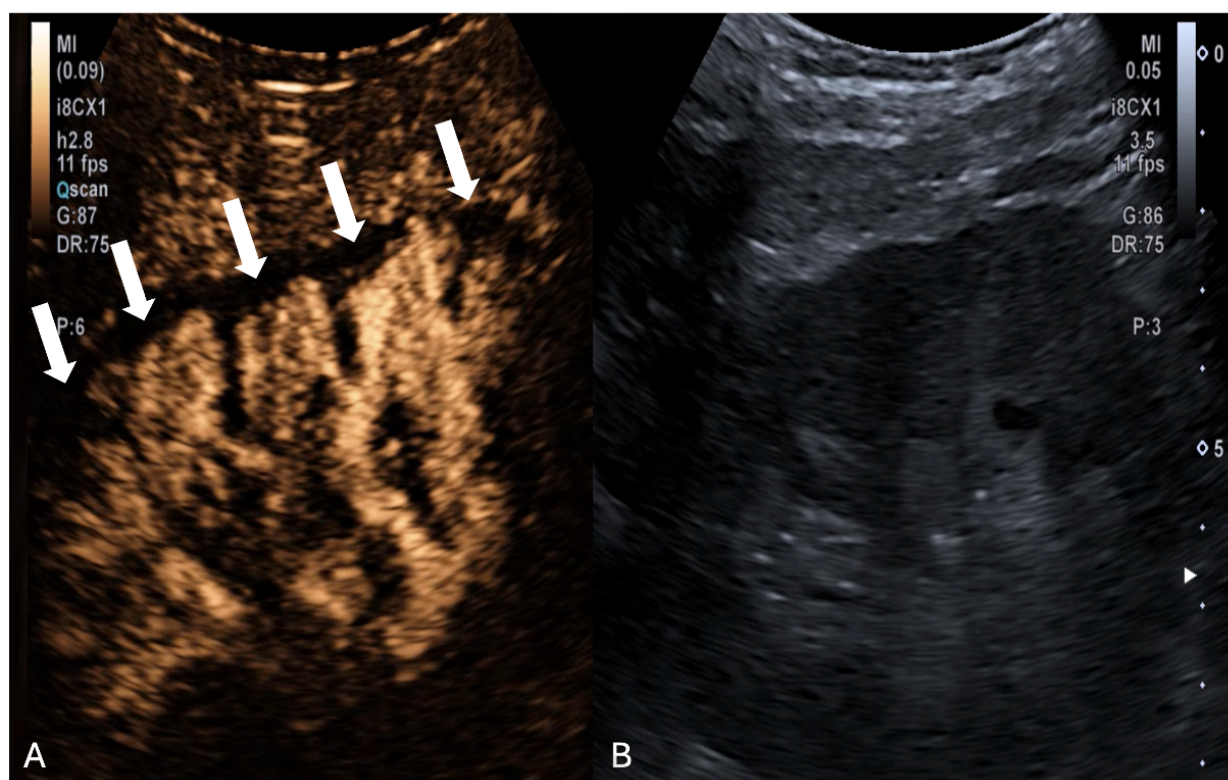
Metric	Definition	How It's Measured	Interpretation*
Time to Peak (TTP)	Time from contrast administration to peak intensity.	From the time-intensity curve (TIC); time at peak on the x-axis.	Typical normal range 10–20 s (cortex), often 7–14 s Prolonged TTP (≥ 30 s) suggests delayed perfusion (e.g., arterial stenosis, rejection).
Peak Intensity (PI)	Maximum intensity of contrast in the region of interest.	Highest value on the TIC.	Low PI (marked reduction >25-30% from baseline) indicates reduced blood flow, ischemia, or vascular occlusion.
Area Under Curve (AUC)	Total amount of contrast-enhanced blood flow over time.	Calculated by integrating the area under the TIC.	A lower AUC can indicate perfusion deficits (e.g., ischemia, rejection).
Wash-in Rate	Rate at which contrast enters the region of interest.	Slope of the ascending portion of the TIC.	Slower wash-in (20-40% decrease) suggests arterial stenosis or delayed perfusion which may be related to microvascular obstruction.
Wash-out Time	Time taken for contrast to leave the region.	From peak intensity to baseline.	Prolonged wash-out indicates venous outflow obstruction (e.g., venous thrombosis).
Rise Time (RT)	Time from initial contrast detection to peak intensity.	From the start of enhancement to peak on the TIC.	Delayed RT can indicate impaired arterial inflow, such as stenosis, or rejection.
Mean Transit Time (MTT)	Average time for contrast to pass through the region.	Average time from the TIC.	Typical normal range 5–10 s Increased MTT (>10-15s) suggests slower perfusion, indicating ischemia or rejection. Short MTT (<5s) can indicate abnormally rapid transit or shunting
Relative Blood Volume (RBV)	Estimated blood volume in the tissue based on contrast enhancement.	Derived from TIC and AUC.	Reduced RBV (>30%) indicates vascular compromise or rejection.
Enhancement Ratio (ER)	Ratio of contrast enhancement in target area to reference area.	PI in target area divided by PI in a reference area.	Abnormal ER (<0.8-0.9) suggests perfusion mismatch due to stenosis or rejection.

Perfusion Index (PID)	Overall perfusion index based on intensity and duration of enhancement.	Combined measure of wash-in, PID, and wash-out phases.	Low PID values suggest compromised vascularization, often seen in rejection.
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*TTP: Time to Peak; PI: Peak Intensity; AUC: Area Under Curve; RT: Rise Time; MTT: Mean Transit Time; RBV: Relative Blood Volume; ER: Enhancement Ratio; TIC: Time-Intensity Curve PID: Perfusion Index. *Ranges can vary with patient factors (cardiac output, heart rate, blood pressure) and equipment settings.*







Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: